

		In the formula.
23	B	P.d. across $5.0\ \Omega$ resistor is proportional to the balanced length, i.e. $5.0\ \Omega = k (12.5\text{ cm})$ Resistor across Q and R is $10.0\ \Omega$. Hence the balanced length should be doubled that of $5.0\ \Omega$, or 25.0 cm.

		Resistor across Q and S is $(10.0 + 15.0 =) 25.0 \, \Omega$. Hence the balanced length should be 5 times that of $5.0 \, \Omega$, or 62.5 cm. Resistor across P and R is $(5.00 + 10.0 =) 15.0 \, \Omega$. Hence the balanced length should be 3 times that of $5.0 \, \Omega$, or 37.5 cm.
24	A	Magnetic force = BIL